REPORT

Faculty of Information and Communication Technology

PROG101: Principles of Programming Logic and Design

1. PROBLEM DESCRIPTION & BACKGROUND

In off-grid communities across Sierra Leone (such as rural areas in the Port Loko and Tonkolili districts), access to clean, reliable electrical power remains a major challenge. While solar mini-grids have been installed in select villages, community operators face significant technical hurdles in managing daily electricity allocation, tracking prepaid energy credits, and preventing total system shutdowns caused by uncontrolled power overdraw.

This problem affects rural household residents, local market traders, health clinic staff, and local energy maintenance technicians. Commercial smart-metering software is prohibitively expensive and requires reliable 4G cellular infrastructure, which is absent in remote off-grid locations. Without basic monitoring logic to control distribution, solar battery storage units suffer severe deep-discharge damage, drastically shortening system lifespans and leaving communities in darkness.

2. COMMUNITY IMPACT & SDG ALIGNMENT

This project aligns directly with Sustainable Development Goal 7 (SDG 7): Affordable and Clean Energy. Specifically, it targets UN SDG Target 7.1 (ensuring universal access to affordable and reliable modern energy) and Target 7.2 (increasing the share of renewable energy reliance).

By introducing an offline-first management logic system, mini-grid operators can easily log household power consumption, calculate remaining energy balances (in kWh), and automatically restrict power delivery when credits run out. The beneficiaries include rural families who gain predictable energy access, microgrid operators who avoid costly equipment replacement, and local businesses that can operate dependably. Long-term sustainability is achieved because automated balance checks prevent battery degradation, keeping infrastructure functional without ongoing licensing fees.

3. PROPOSED SOLUTION & COMPUTATIONAL THINKING PROCESS

The proposed solution applies the four core pillars of Computational Thinking:

> **1. Decomposition:** Breaking the overall energy management system into discrete logic tasks: Household Registration, Energy Credit Top-Up, Consumption Logging, and System Status Reporting.

> **2. Pattern Recognition:** Identifying that power allocation follows consistent mathematical rules (Remaining Credit = Current Balance - Consumed Power).

> **3. Abstraction:** Removing administrative complexity so that non-technical village operators simply interact with numbered menu choices while background calculations run automatically.

> **4. Algorithm Design:** Building clear step-by-step control logic using decision structures (if-else), repetition loops (while), and modular subroutines.

4. STEP-BY-STEP ALGORITHM DESIGN

1. **Step 1:** Start the program and set the maximum daily grid capacity limit.
2. **Step 2:** Display the main menu options to the system operator (1: Register Household, 2: Top-Up Credits, 3: Log Usage, 4: System Status, 5: Exit).
3. **Step 3:** If Option 1 is chosen, record household details and generate an Account ID.
4. **Step 4:** If Option 2 is chosen, accept Account ID and credit amount, then add it to the balance.
5. **Step 5:** If Option 3 is chosen, accept Account ID and energy consumed. If consumption is less than or equal to balance, deduct usage; otherwise, reject request and alert operator.
6. **Step 6:** If Option 4 is chosen, display grid totals for total power generated vs. consumed.

7. **Step 7:** If Option 5 is chosen, terminate program execution loop.

7. FUNCTIONAL MODULES & FUNCTIONS

The solution uses modular program architecture to segregate responsibilities:

- **RegisterHousehold():** Manages memory arrays, validates system limits, and assigns unique 4-digit Account IDs (7001, 7002, etc.).
- **TopUpEnergy():** Increases individual user accounts by adding purchased kWh units.
- **LogUsage():** Executes decision logic (if-else) to evaluate usage requests against available balances, protecting the microgrid from overloading.
- **DisplayStatus():** Compiles real-time metrics showing aggregate solar power generation versus energy consumed across all registered users.

9. OPEN SOURCE & DIGITAL PUBLIC GOODS (DPG) JUSTIFICATION

Releasing this microgrid control logic under an open-source license (such as MIT) ensures that local community organizations, NGOs, and Sierra Leonean developers can inspect, modify, and improve the software at zero cost.

This approach builds national software development capacity, removes financial barriers to rural electrification, and aligns with Digital Public Goods (DPG) standards by delivering accessible technological benefit for public good.